

Clustering

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Clustering

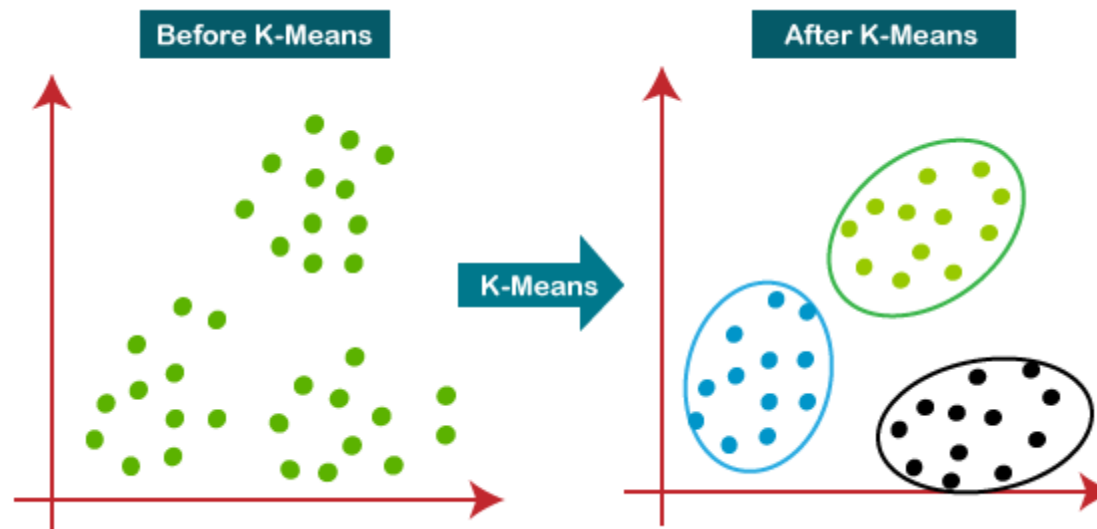
- Clustering is the task of grouping a set of objects such that similar objects end up in the same group and dissimilar objects are separated into different groups.
- Clustering is an unsupervised learning problem; i.e., there are no class labels to train the model.
- **Ex:** Suppose, the head of a rental store wishes to understand preferences of costumers to scale up the business.
- It is not possible to look at details of each costumer and devise a unique business strategy for each one of them?
- But, all the costumers can be grouped into say 10 groups based on their purchasing habits and use a separate strategy for costumers in each of these 10 groups.
- This is what we call clustering.

Applications of Clustering

- **Marketing :** It can be used to characterize & discover customer segments for marketing purposes.
- **Biology :** It can be used for classification among different species of plants and animals.
- **Libraries :** It is used in clustering different books on the basis of topics and information.
- **Insurance :** It is used to acknowledge the customers, their policies and identifying the frauds.
- **City Planning:** It is used to make groups of houses and to study their values based on their geographical locations and other factors present.
- **Earthquake studies:** By learning the earthquake-affected areas we can determine the dangerous zones.

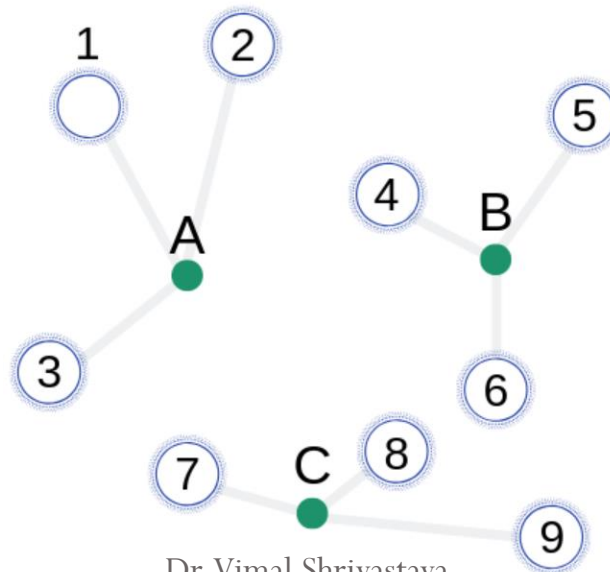
K-Means Clustering

- It is an Unsupervised Learning algorithm.
- It comes under the Centroid-based Clustering.
- The k-means clustering algorithm assigns data points to clusters, by finding the mean distance between data points.



Algorithm

- **Step1:** First, we need to specify the number of clusters, K , need to be generated by this algorithm. (It assumes that the number of clusters are already known.)
- **Step2:** Randomly select K data points and assign each data point to a cluster.
- **Step3** – Now, compute the cluster centroids (based on the means of the data points) for each cluster.



- **Step 4** – Keep iterating the following until we find optimal centroid which is the assignment of data points to the clusters that are not changing any more.
- **4.1** – First, the sum of squared distance between data points and centroids would be computed.
- Euclidean Distance:

$$\begin{aligned}d(\mathbf{p}, \mathbf{q}) &= d(\mathbf{q}, \mathbf{p}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \cdots + (q_n - p_n)^2} \\&= \sqrt{\sum_{i=1}^n (q_i - p_i)^2}.\end{aligned}$$

- **4.2** – Now, we have to assign each data point to the cluster that is closer than other cluster (centroid).
- **4.3** – At last compute the centroids for the clusters by taking the average of all data points of that cluster.

Distance Measures

- The choice of distance measures is a critical step in clustering. It defines how the similarity of two elements (x, y) is calculated and it will influence the shape of the clusters.
- The classical methods for distance measures are *Euclidean* and *Manhattan distances*, which are defined as follow:

- *Euclidean distance*:

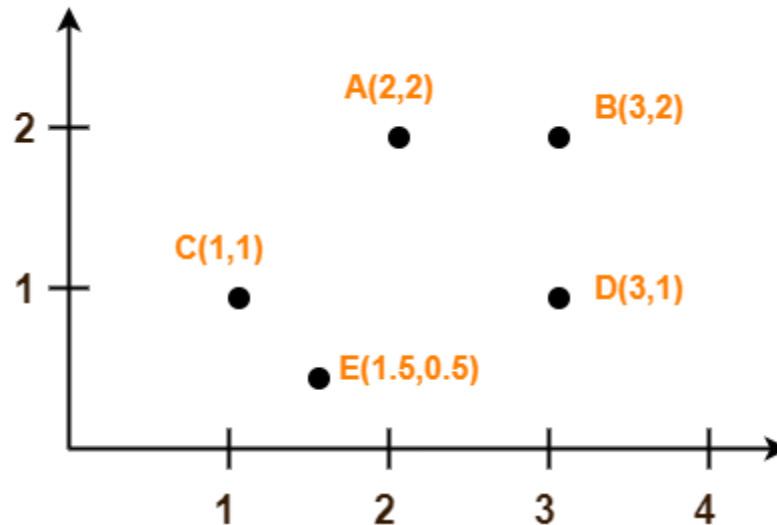
$$d_{euc}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

- *Manhattan distance*:

$$d_{man}(x, y) = \sum_{i=1}^n |(x_i - y_i)|$$

Problem: 01

- Use K-Means Algorithm to create two clusters.



- **Solution:**

Assume $A(2, 2)$ and $C(1, 1)$ are centers of the two clusters.

Iteration-01:

- We calculate the distance of each point from each of the center of the two clusters.
- The distance is calculated by using the Euclidean distance formula.

Calculating Distance Between A(2, 2) and B(3, 2)-

$$\begin{aligned} P(A, B) &= \text{sqrt} [(x_2 - x_1)^2 + (y_2 - y_1)^2] \\ &= \text{sqrt} [(3 - 2)^2 + (2 - 2)^2] \\ &= \text{sqrt} [1 + 0] \\ &= 1 \end{aligned}$$

- In the similar manner, we calculate the distance of other points from each of the center of the two clusters.

Given Points	Distance from center (2, 2) of Cluster-01	Distance from center (1, 1) of Cluster-02	Point belongs to Cluster
A(2, 2)	0	1.41	C1
B(3, 2)	1	2.24	C1
C(1, 1)	1.41	0	C2
D(3, 1)	1.41	2	C1
E(1.5, 0.5)	1.58	0.71	C2

From here, New clusters are-

Cluster-01:

First cluster contains points-

- A(2, 2)
- B(3, 2)
- D(3, 1)

Cluster-02:

Second cluster contains points-

- C(1, 1)
- E(1.5, 0.5)

Now,

- We re-compute the new cluster clusters.
- The new cluster center is computed by taking mean of all the points contained in that cluster.

For Cluster-01:

Center of Cluster-01

$$\begin{aligned} &= ((2 + 3 + 3)/3, (2 + 2 + 1)/3) \\ &= (2.67, 1.67) \end{aligned}$$

For Cluster-02:

Center of Cluster-02

$$\begin{aligned} &= ((1 + 1.5)/2, (1 + 0.5)/2) \\ &= (1.25, 0.75) \end{aligned}$$

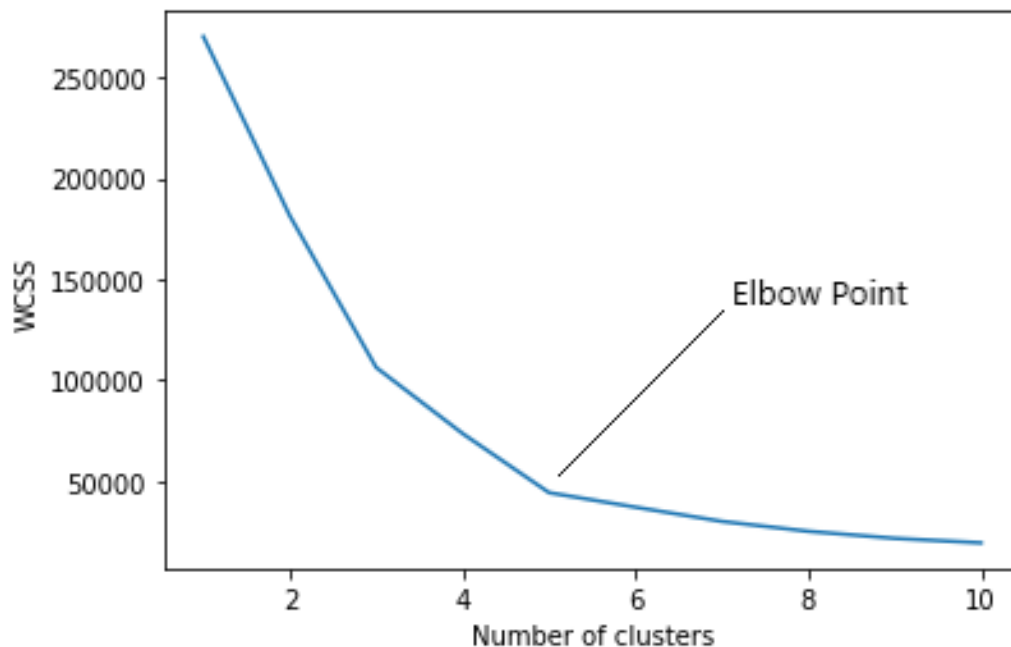
- This is completion of Iteration-01.
- Next, we go to iteration-02, iteration-03 and so on until the centers do not change anymore.

Pros and Cons

- **Pros:**
 - Fast and efficient.
 - Works on unlabeled numerical data.
 - Iterative technique.
- **Cons:**
 - Must understand the context of your data well.
 - Have to choose your own k value.
 - Lots of repetition.
 - Does not perform well when outliers are present.

Elbow Method: Finding the Optimal Value of 'K'

- The elbow method is a graphical representation of finding the optimal 'K' in a K-means clustering.
- It works by finding WCSS (Within-Cluster Sum of Square).
- WCSS is the sum of the squared distance between each point and the centroid in a cluster.

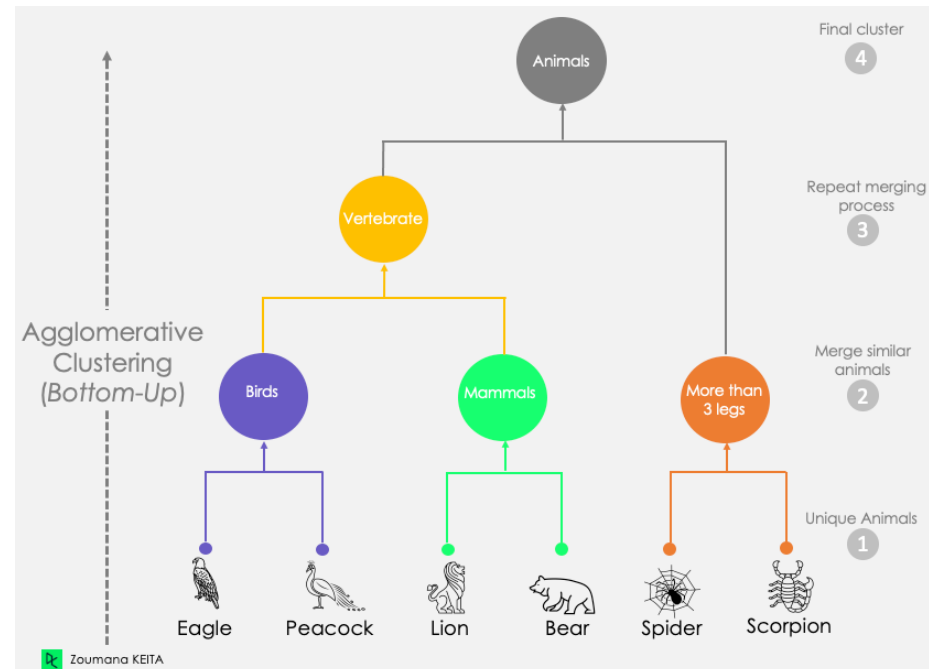


Hierarchical Clustering

- Clusters are visually represented in a hierarchical tree called a dendrogram.
- There are two main types of hierarchical clustering:
- **Agglomerative:**
 - Initially, each object is considered to be its own cluster.
 - According to a particular procedure, the clusters are then merged step by step until a single cluster remains.
 - At the end of the cluster merging process, a cluster containing all the elements will be formed.

Example

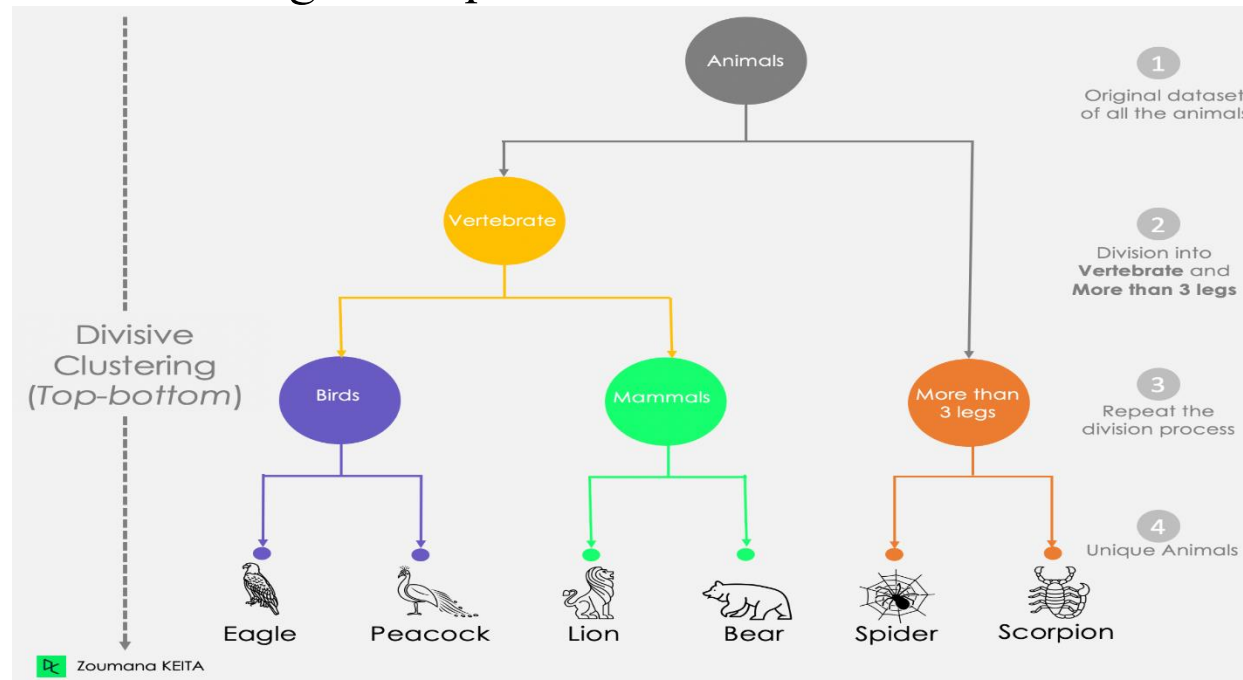
- We start by considering each animal to be its unique cluster.
- Then we generate three different clusters from those unique animals based on their similarities:
 - Birds: Eagle and Peacock
 - Mammals: Lion and Bear
 - More than three leg animals: Spider and Scorpion.
- We repeat the merging process to create the vertebrate cluster by combining the two most similar clusters: Birds and Mammals.
- After this step, the remaining two clusters, Vertebrate and More than three legs, are merged to create a single Animals cluster.



Dendrogram of Agglomerative Clustering Approach

- **Divisive:**
- Divisive method is the opposite of the Agglomerative method.
- Initially, all objects are considered in a single cluster.
- Then the division process is performed step by step until each object forms a different cluster.
- Between Agglomerative and Divisive clustering, Agglomerative clustering is generally the preferred method.

- **Example:**
- We notice that the whole animal dataset is considered as a single bloc.
- Then, we divide that block into two clusters: Vertebrate and More than 3 legs.
- The division process is iteratively applied to the previously created clusters until we get unique animals.



Dendrogram of Divisive Clustering Approach

Agglomerative Hierarchical Clustering

- Step 1: Compute the proximity matrix using a particular distance metric.
- Step 2: Each data point is assigned to a cluster.
- Step 3: Merge the clusters based on a metric for the similarity between clusters.
- Step 4: Update the distance matrix.
- Step 5: Repeat Step 3 and Step 4 until only a single cluster remains.

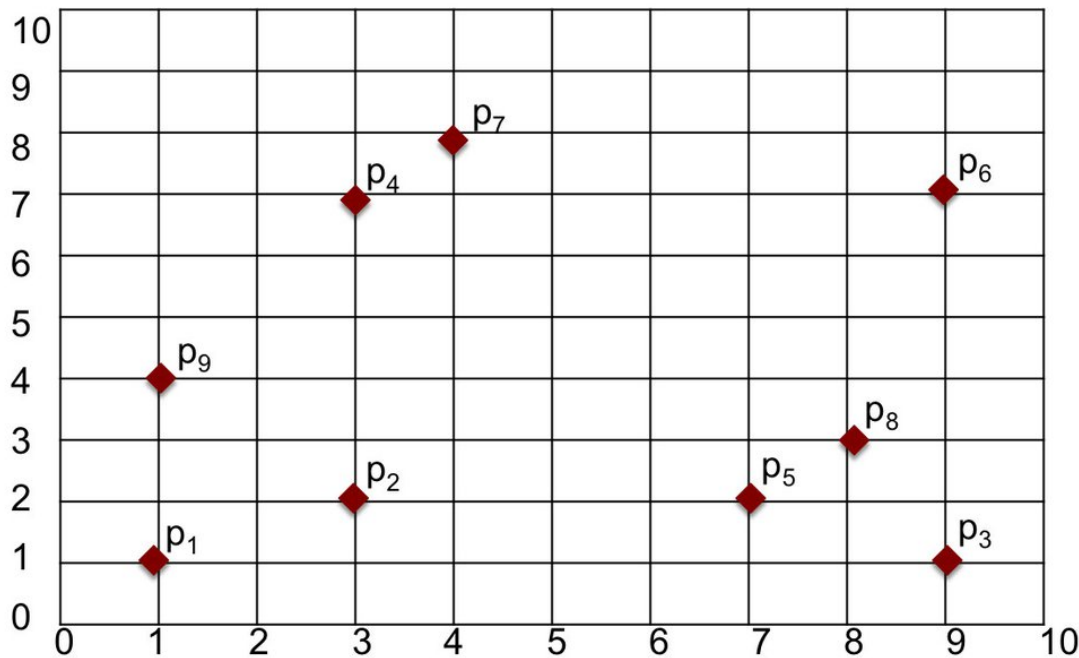
Computing a Proximity Matrix

- The values of the matrix are calculated by applying a Euclidean distance function between each pair of objects.
- Here, $d(p_i, p_j)$ represent the distance values between p_i and p_j .

	p_1	p_2	p_3	$\dots$	p_n
p_1	$d(p_1, p_1)$	$d(p_1, p_2)$	$d(p_1, p_3)$	$\dots$	$d(p_1, p_n)$
p_2	$d(p_2, p_1)$	$d(p_2, p_2)$	$d(p_2, p_3)$	$\dots$	$d(p_2, p_n)$
p_3	$d(p_3, p_1)$	$d(p_3, p_2)$	$d(p_3, p_3)$	$\dots$	$d(p_3, p_n)$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
p_n	$d(p_n, p_1)$	$d(p_n, p_2)$	$d(p_n, p_3)$	$\dots$	$d(p_n, p_n)$

Similarity Between Clusters

- The main question is how to calculate the distance between clusters and update the proximity matrix.
- **There are different approaches:**
- Let's take a small sample dataset containing just nine two-dimensional points as shown in Figure 1:



Point	x	y
p_1	1	1
p_2	3	2
p_3	9	1
p_4	3	7
p_5	7	2
p_6	9	7
p_7	4	8
p_8	8	3
p_9	1	4

Figure 1

- Suppose we have two clusters in the sample data set, as shown in Figure 2.

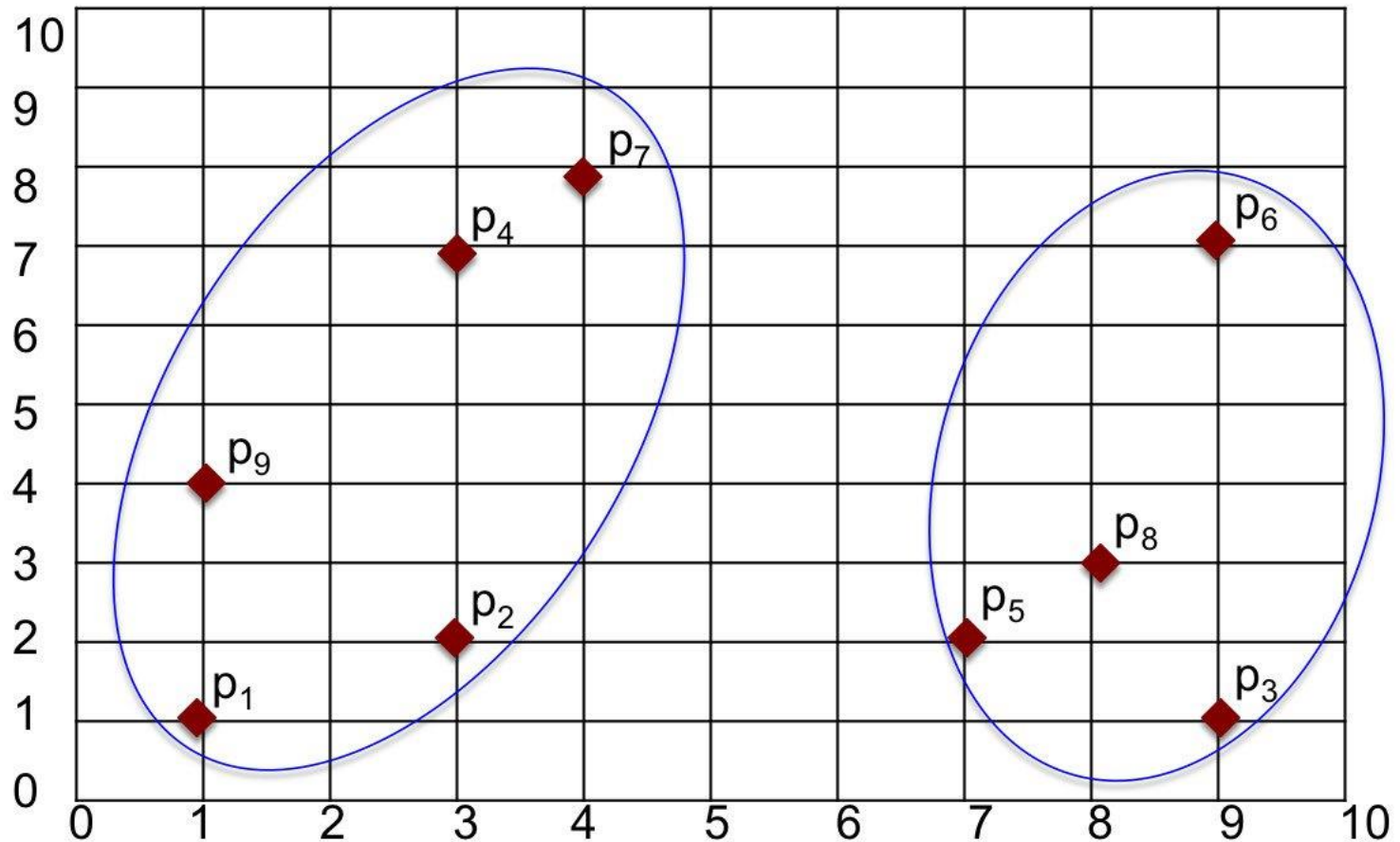


Figure 2

Min (Single) Linkage method

- One way to measure the distance between clusters is to find the minimum distance between points in those clusters.
- That is, we can find the point in the first cluster nearest to a point in the other cluster and calculate the distance between those points.
- In Figure 2, the closest points are p_2 in one cluster and p_5 in the other.
- The distance between those points, and hence the distance between clusters, is found as $d(p_2, p_5)=4$.

Distance (C1, C2) = Min { $d(i, j)$, where item i is within C1, and item j is within C2 }

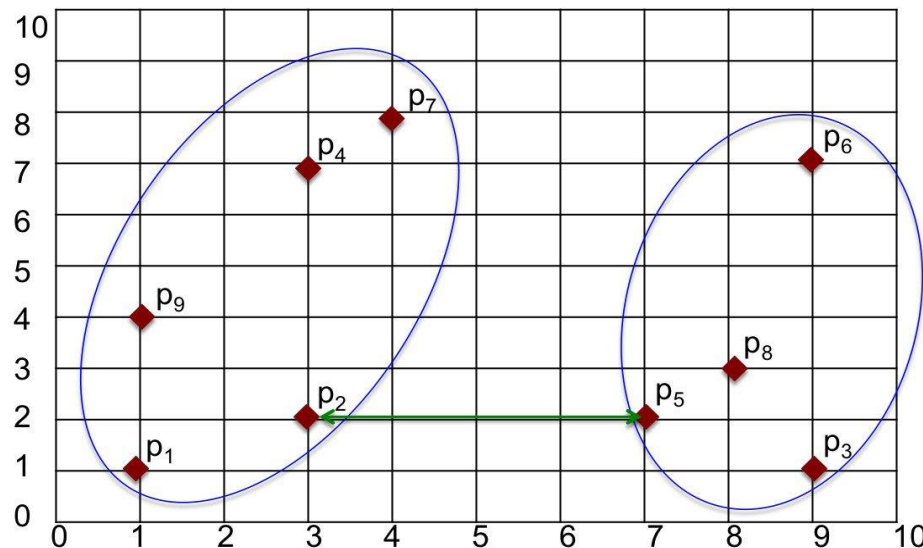


Figure 3: Min Linkage Method

Max (Complete) Linkage

- Another way to measure the distance is to find the maximum distance between points in two clusters.
- We can find the points in each cluster that are furthest away from each other and calculate the distance between those points.
- In Figure 3, the maximum distance is between p_1 and p_6
- Distance between those two points, and hence the distance between clusters, is found as $d(p_1, p_6)=10$.

Distance (C1, C2) = Max { $d(i, j)$, where item i is within C1, and item j is within C2 }

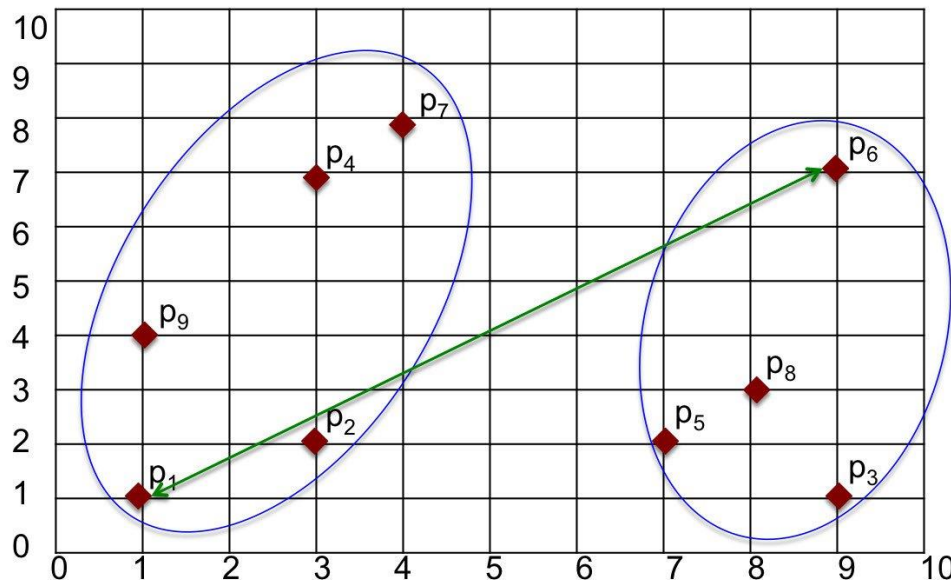


Figure 4: Max Linkage Method

Centroid Linkage

- The Centroid method defines the distance between clusters as being the distance between their centers/centroids.
- After calculating the centroid for each cluster, the distance between those centroids is computed using a distance function.

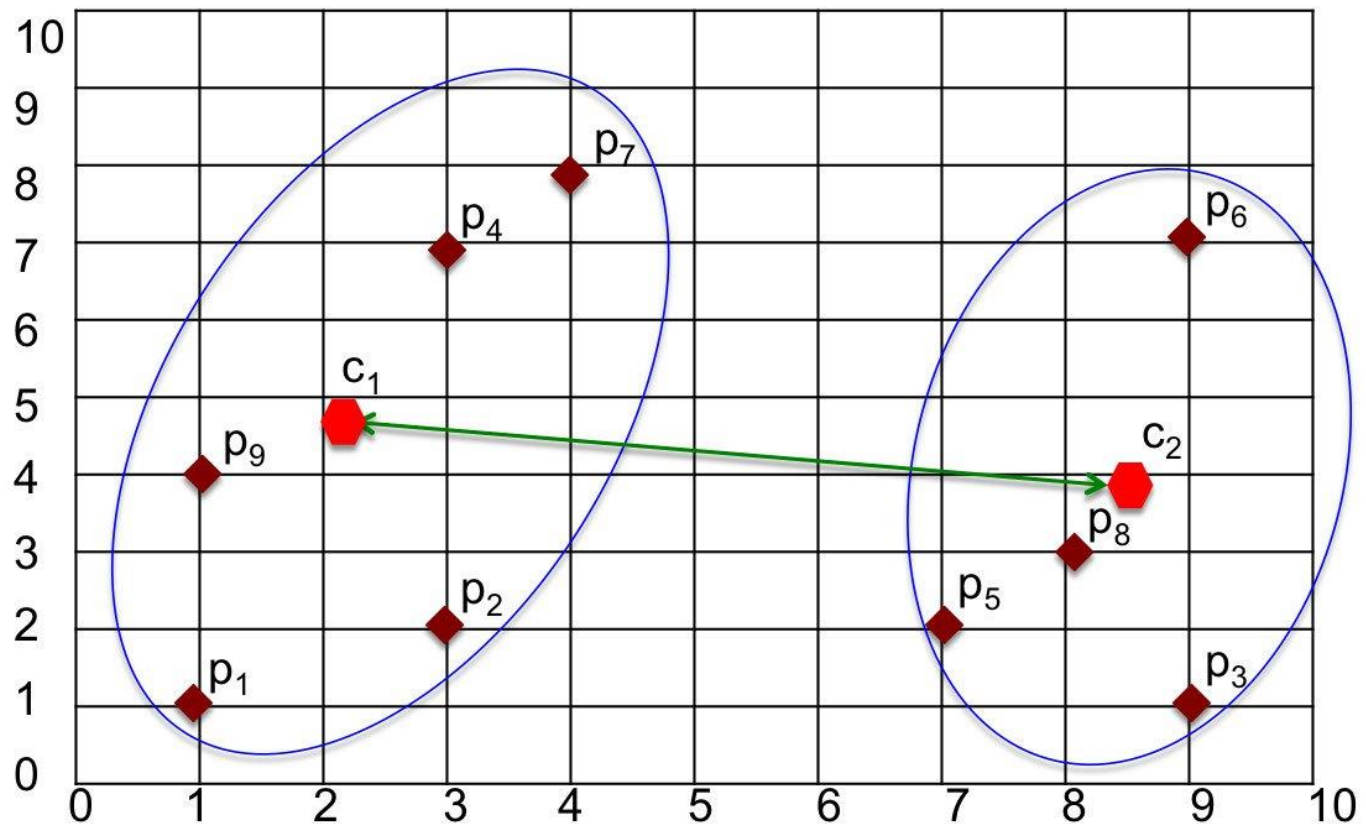


Figure 5: Centroid Linkage Method

Average Linkage

- The Average method defines the distance between clusters as the average pairwise distance among all pairs of points in the clusters.
- For simplicity, only some of the lines connecting pairs of points are shown in Figure 6.

$$\text{Distance}(C1, C2) = \text{Sum}\{d(i, j)\} / \text{Total Number of distances}$$

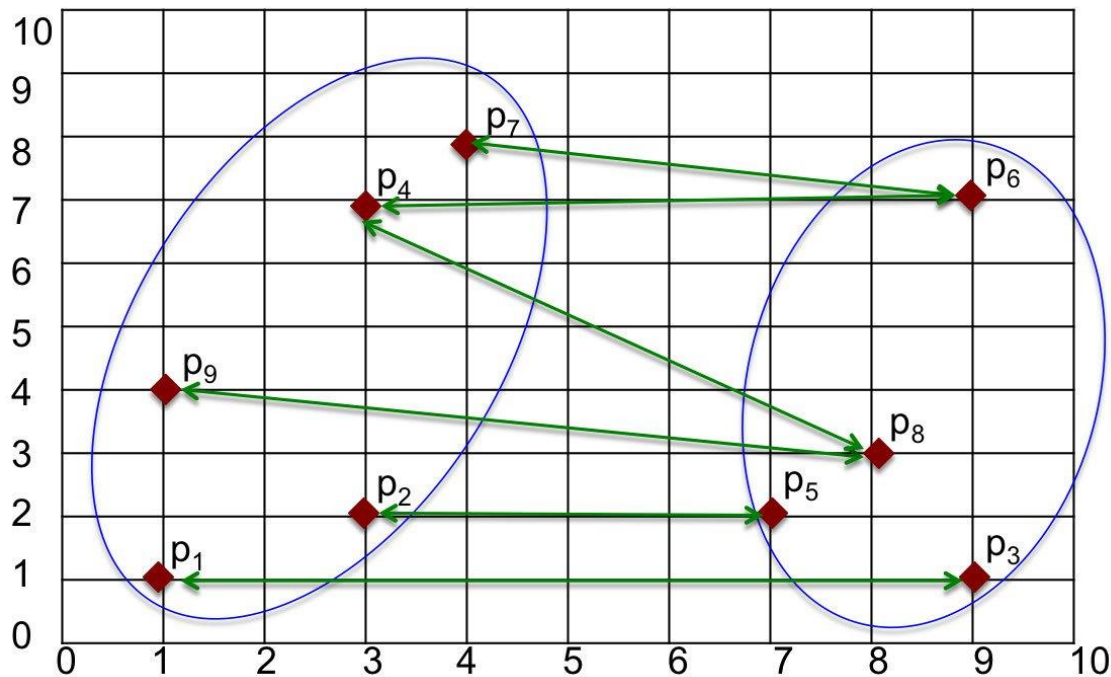


Figure 6: Average Linkage Method

Ward Linkage

- The Ward approach analyzes the variance of the clusters rather than measuring distances directly, minimizing the variance between clusters.
- It attempts to minimize the sum of the squared distances of the points from the cluster centers.
- Compared to the distance-based measures described above, the Ward method is less susceptible to noise and outliers. Therefore, Ward's method is preferred more than others.

Key Benefits:

- There is no need to pre-specify the number of clusters. Instead, the dendrogram can be cut at the appropriate level to obtain the desired number of clusters.
- Data is easily summarized/organized into a hierarchy using dendrograms. Dendrograms make it easy to examine and interpret clusters.

Drawbacks:

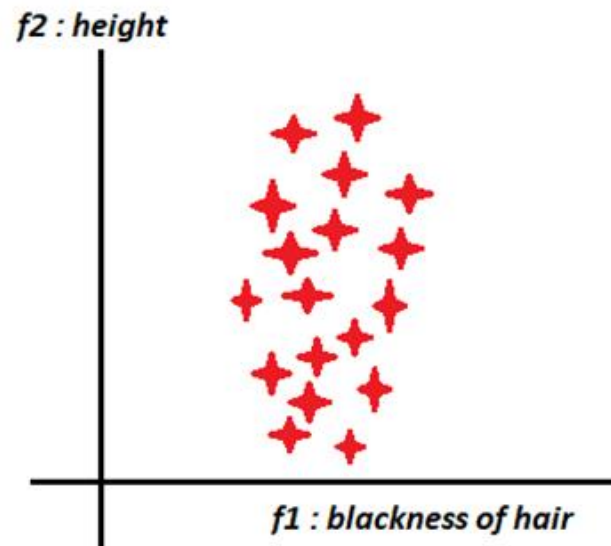
- The computational cost can be high, especially for large datasets.
- The results can be sensitive to the linkage criterion, and distance metric used.

Applications:

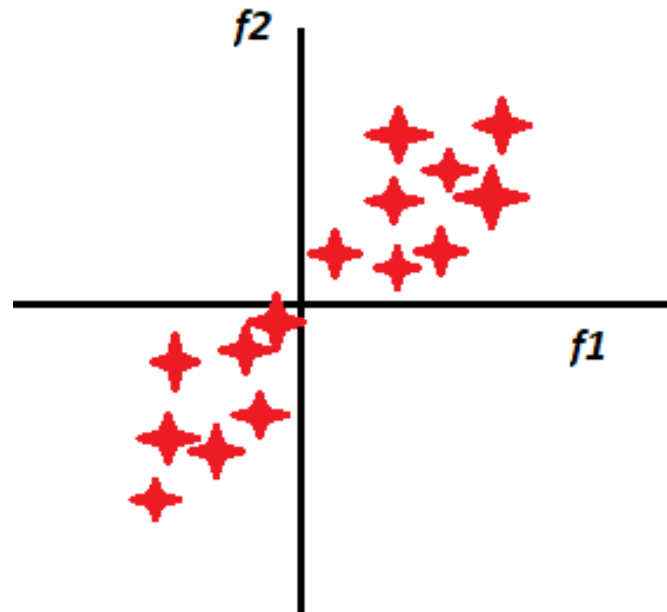
- **Business:** dividing customers into segments or forming a hierarchy of employees based on salary.
- **Image processing:** grouping handwritten characters in text recognition based on the similarity of the character shapes.
- **Information Retrieval:** categorizing search results based on the query.

Principal Component Analysis (PCA)

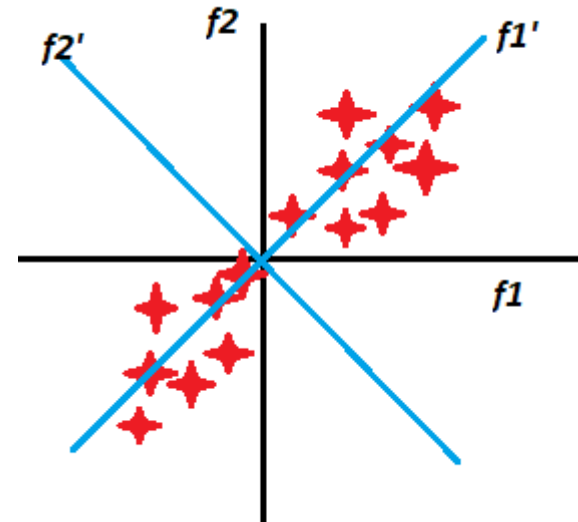
- PCA, is a dimensionality-reduction method that is often used to reduce the dimensionality of large dataset.
- It transforms a large set of variables into a smaller one that still contains most of the information.



- To convert this 2D data into 1D data, which feature should be selected?
- We should obviously select $f2$ (height), as it can be seen that blackness of hair doesn't have much spread/information in it. So even removing it and representing all points just in terms of height would suffice.
- Now, what about the following data. Which feature should be selected?



- Both of the features $f1$ and $f2$ have nearly equal spread, so we come up with new features as :



- Now , it has become similar to previous problem.
- Feature $f1'$ has a lot of spread/information while $f2'$ doesn't . So we select only $f1'$ and drop $f2'$.
- Summary: Select those features along which we get **high variance** and drop those along which we get less variance .

Algorithm

- **STEP 1: Create mean center data:**
- We will do mean centering by subtracting mean from all features.

- **STEP 2: COVARIANCE MATRIX COMPUTATION**

- The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other or in other words, to see if there is any relationship between them.
- The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions).
- For example, for a 3-dimensional data set with 3 variables x , y , and z , the covariance matrix is a 3×3 matrix of this form:

$$\begin{bmatrix} Cov(x, x) & Cov(x, y) & Cov(x, z) \\ Cov(y, x) & Cov(y, y) & Cov(y, z) \\ Cov(z, x) & Cov(z, y) & Cov(z, z) \end{bmatrix}$$

- It's actually the sign of the covariance that matters:
 - if positive then : the two variables increase or decrease together (correlated)
 - if negative then : One increases when the other decreases (Inversely correlated)

- **Step 3: Compute Eigenvectors and Eigenvalues of Covariance Matrix to Identify the Principal Components**
- Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data.
- **Principal Components:**
- Principal components are constructed in such a manner that the first principal component accounts for the largest possible variance in the data set.
- **STEP 4: Feature Vector**
- Feature vector is simply a matrix that has as columns the eigenvectors of the components that we decide to keep. This makes it the first step towards dimensionality reduction, because if we choose to keep only p eigenvectors (components) out of n , the final data set will have only p dimensions.

PCA Algorithm

- The steps involved in PCA Algorithm are as follows:
- Step-01: Get the data.
- Step-02: Compute the mean vector (μ).
- Step-03: Subtract mean from the given data.
- Step-04: Calculate the covariance matrix.
- Step-05: Calculate the Eigen vectors and Eigen values of the covariance matrix.
- Step-06: Choose components and form a feature vector.
- Step-07: Derive the new data set.

Problem-01

- Given data = $\{ 2, 3, 4, 5, 6, 7 ; 1, 5, 3, 6, 7, 8 \}$.
- Compute the principal component using PCA Algorithm.

OR

- Consider the two dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6, 7), (7, 8).
- Compute the principal component using PCA Algorithm.

OR

- Compute the principal component of following data-
- CLASS 1
 - $X = 2, 3, 4$
 - $Y = 1, 5, 3$
- CLASS 2
 - $X = 5, 6, 7$
 - $Y = 6, 7, 8$

Source: <https://www.gatevidyalay.com/tag/principal-component-analysis-numerical-example/>

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Solution

- **Step-01:** Get the data.
- The given feature vectors are:
 - $x_1 = (2, 1)$
 - $x_2 = (3, 5)$
 - $x_3 = (4, 3)$
 - $x_4 = (5, 6)$
 - $x_5 = (6, 7)$
 - $x_6 = (7, 8)$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

- **Step-02:** Calculate the mean vector (μ).
- Mean vector (μ)
 $= ((2 + 3 + 4 + 5 + 6 + 7) / 6, (1 + 5 + 3 + 6 + 7 + 8) / 6)$
 $= (4.5, 5)$

- Thus,

$$\text{Mean vector } (\mu) = \begin{bmatrix} 4.5 \\ 5 \end{bmatrix}$$

- **Step-03:** Subtract mean vector (μ) from the given feature vectors.
 - $x_1 - \mu = (2 - 4.5, 1 - 5) = (-2.5, -4)$
 - $x_2 - \mu = (3 - 4.5, 5 - 5) = (-1.5, 0)$
 - $x_3 - \mu = (4 - 4.5, 3 - 5) = (-0.5, -2)$
 - $x_4 - \mu = (5 - 4.5, 6 - 5) = (0.5, 1)$
 - $x_5 - \mu = (6 - 4.5, 7 - 5) = (1.5, 2)$
 - $x_6 - \mu = (7 - 4.5, 8 - 5) = (2.5, 3)$

- Feature vectors (x_i) after subtracting mean vector (μ) are:

$$\begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 2.5 \\ 3 \end{bmatrix}$$

- **Step-04:** Calculate the covariance matrix.
- Covariance matrix is given by-

$$\text{Covariance Matrix} = \frac{\sum (x_i - \mu)(x_i - \mu)^t}{N-1}$$

$$m_1 = (x_1 - \mu)(x_1 - \mu)^t = \begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$

$$m_2 = (x_2 - \mu)(x_2 - \mu)^t = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$

$$m_3 = (x_3 - \mu)(x_3 - \mu)^t = \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$

$$m_4 = (x_4 - \mu)(x_4 - \mu)^t = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$m_5 = (x_5 - \mu)(x_5 - \mu)^t = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$m_6 = (x_6 - \mu)(x_6 - \mu)^t = \begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

- Covariance matrix

$$= (m_1 + m_2 + m_3 + m_4 + m_5 + m_6) / 6$$

- On adding the above matrices and dividing by 6, we get-

$$\begin{aligned} \text{Covariance Matrix} &= \frac{1}{5} \begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix} \\ &= \begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} \end{aligned}$$

- **Step-05:** Calculate the Eigen values and Eigen vectors of the covariance matrix.
- λ is an Eigen value for a matrix M if it is a solution of the characteristic equation $|M - \lambda I| = 0$.
- So, we have:
$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

- From here,
- $(3.5 - \lambda)(6.8 - \lambda) - (4.4 \times 4.4) = 0$

- Solving this quadratic equation, we get $\lambda = 0.45, 9.85$
- Thus, two Eigen values are $\lambda_1 = 0.45$ and $\lambda_2 = 9.85$.
- Clearly, the first Eigen value is very small compared to the second Eigen value.
- So, the first Eigen vector can be left out.
- Eigen vector corresponding to the greatest Eigen value is the principal component for the given data set.
- So. we find the Eigen vector corresponding to Eigen value λ_2 .

- We use the following equation to find the Eigen vector:

$$MX = \lambda X$$

- where-

- M = Covariance Matrix
- X = Eigen vector
- λ = Eigen value

- Substituting the values in the above equation, we get:

$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = 9.85 \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

- Solving these, we get-
- $3.5X_1 + 4.4X_2 = 9.85X_1$
- $4.4X_1 + 6.8X_2 = 9.85X_2$

$$X_1 = 0.69 X_2$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 0.69 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} \frac{0.69}{\sqrt{(0.69^2 + 1^2)}} \\ \frac{1}{\sqrt{(0.69^2 + 1^2)}} \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 0.57 \\ 0.82 \end{bmatrix}$$

- From (2), the Eigen vector is:

$$\text{Eigen Vector : } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.57 \\ 0.82 \end{bmatrix}$$

- **Transform Original Dataset:**

$$Z = XV$$

where,

X = Mean Center Data

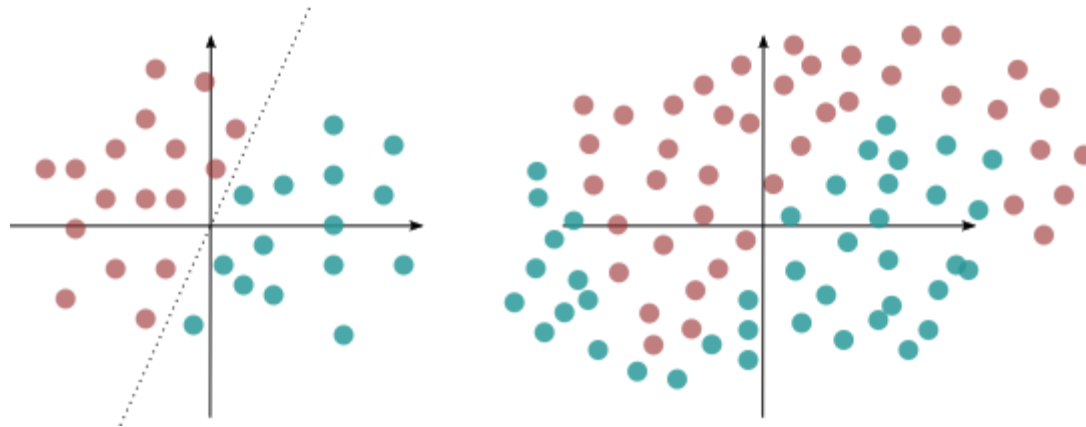
V = Eigen Vector

$$Z = [-4.71, -0.85, -1.93, 1.11, 2.50, 3.89]$$

- **The number of principal components is less than or equal to the number of original variables.**

Kernel PCA

- PCA is a linear algorithm.
- In order to deal with the presence of non-linearity in the data, the technique of kernel PCA was developed.



- Kernel PCA was developed to help with the classification of data whose decision boundaries are described by non-linear function.
- The idea is to go to a higher dimension space in which the decision boundary becomes linear.
- It is achieved using “Kernel Trick” for example Gaussian kernel which is also called ‘RBF’ (radial-basis function).